

CAUSAL INFERENCE PROJECT

Nonlinear Difference-in-Differences

Methods & methodology for assessing marketing interventions

Geminos Team

April 25, 2026

Develop a dataset and methodology for assessing marketing interventions

Background:

Netflix often run marketing interventions to increase user engagement (e.g., recommendations, promotions)

Challenge:

1. users may not be randomly treated
2. user behavior depends on features like age and preferences
3. these features often relate to the outcome in a nonlinear way

Main question:

Can we recover the true effect of the intervention under this nonlinear relationship?

01

Build

Simulate users and their behavior, include non-random treatment assignment and nonlinear effects

02

Model

Apply standard DiD and extend to more flexible models

03

Compare

Compare estimated effects to the true value and evaluate bias

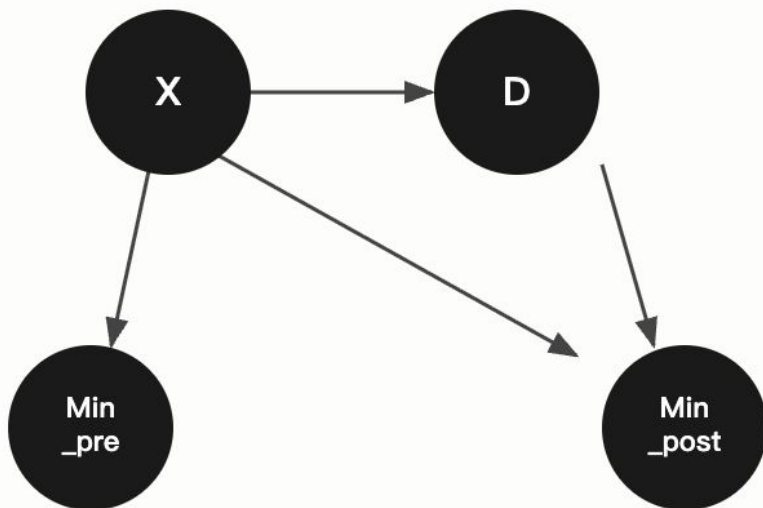
Synthetic Data Setup — What we simulated

We simulated 2,000 users and tracked them before and after an intervention.

01 Age group age User age? Between 20 and 30 Years old	02 Genre preferences g Types of genre? Action, Drama, Comedy, SciFi, Romance.	03 Times of measurement t When we observe the user? Before intervention After intervention	04 Intervention $D \in \{0,1\}$ Receiving Treatment? 0 = No 1 = Yes (only in post period)	05 Minutes viewed Y Outcome Variable: How much time the user watches
---	---	--	---	---

How variables affect treatment and outcome

User characteristics influence both who gets treated and how much they watch.



Treatment is not randomized

User characteristics affect both who gets treated and how much they watch

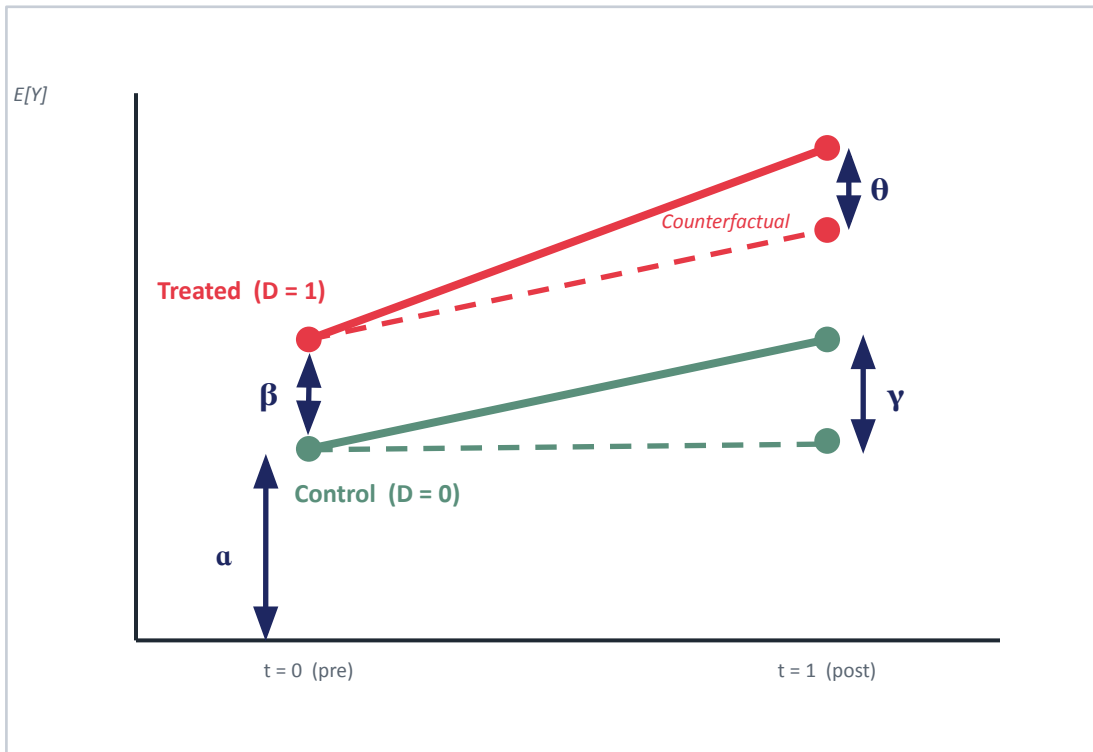
Behavior changes over time

Even without treatment, viewing changes over time and this change depends on the user

What DiD does

DiD compares changes over time to isolate the treatment effect from unobserved temporal confounding

DiD idea: compare changes over time



DID Assumptions:

1 Parallel trends

Without treatment, both groups would change in the same way over time

2 No anticipation

The treatment does not affect viewer behavior before the intervention begins

Two-way fixed effects DID — the linear baseline

$$Y_{it} = \alpha + \beta D_i + \gamma \text{Post}_t + \theta(D_i \times \text{Post}_t) + \theta_x^\top X_i + \varepsilon_{it}$$

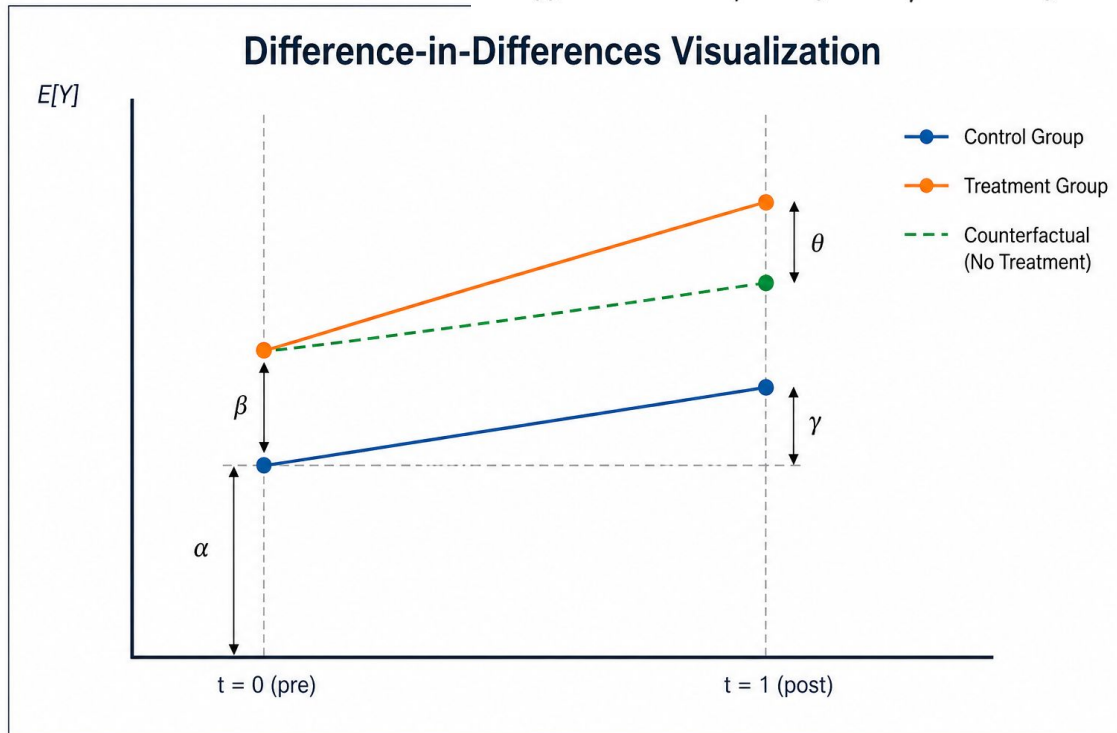
Linear specification — covariates enter additively as $\theta_x' X$

α	β	γ	θ	$\theta_x' X$
Intercept	Group effect	Time effect	DID coefficient	Covariate adjustment
Grand mean of control at $t = 0$	Baseline level gap: treated vs. control at the pre period	Common secular trend — what control's pre→post jump looks like	The ATT — vertical gap between observed treated post and counterfactual	Linear correction for X. Replaced by $f(X)$ in the next step.

Map to the DID plot: α sets the baseline, β raises the treated line, γ tilts both lines, and θ is the wedge between observed treated and the parallel-trend counterfactual.

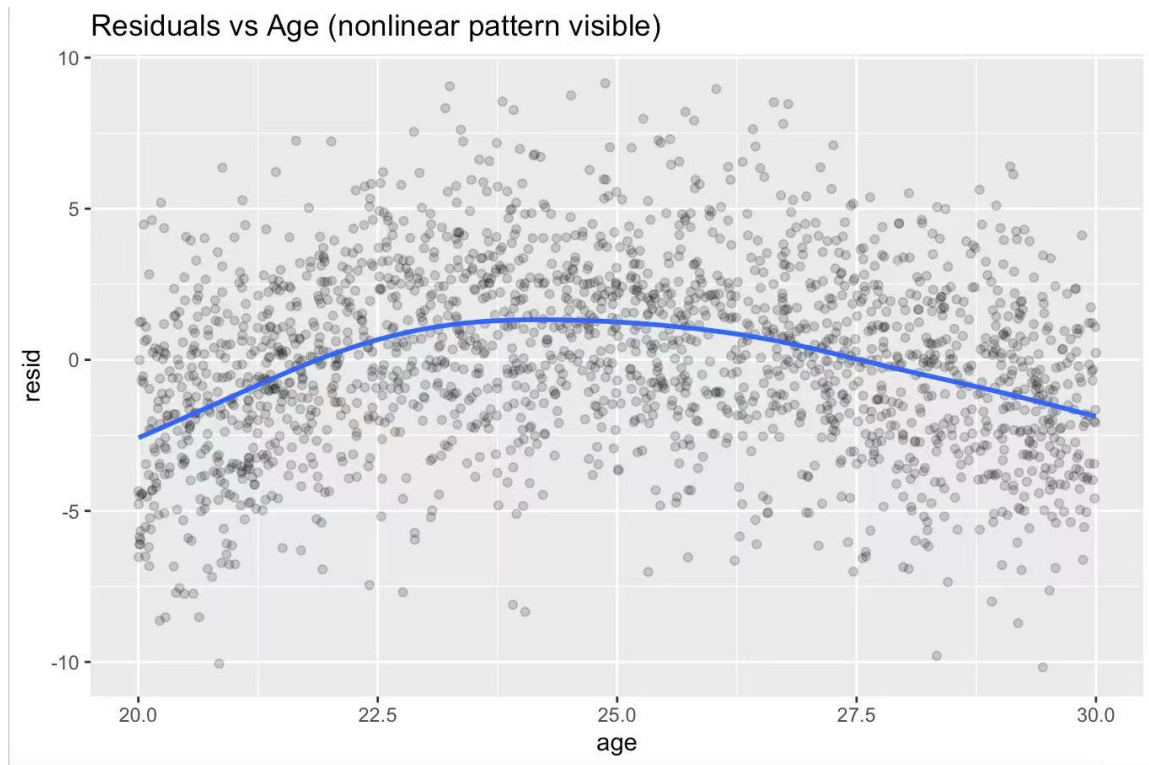
Interpretation of DiD Model Parameters

$$Y_{it} = \alpha + \beta D_i + \gamma \text{Post}_t + \theta(D_i \times \text{Post}_t) + \boldsymbol{\theta}_x^\top X_i + \varepsilon_{it}$$



α : baseline
 β : group difference
 γ : time effect
 θ : treatment effect
 X : covariates (not shown)

Nonlinear Relationship in Covariates



Age might have a nonlinear
effect on viewing time
Linear models may be
misspecified

Replace $\theta_x' X$ with a flexible $f(X)$

Linear (misspecified)

$$\mathbb{E}[\Delta Y_i(0) \mid X_i] = X_i^\top \beta$$

Nonlinear (NCPT)

$$\mathbb{E}[\Delta Y_i(0) \mid X_i] = \mu_0(X_i) = f(X_i)$$

Identification under Nonlinear Conditional Parallel Trends (NCPT)

$$\text{ETT} = \mathbb{E}[\Delta Y_i \mid A_i = 1] - \mathbb{E}[\mu_0(X_i) \mid A_i = 1], \quad \mu_0(x) = \mathbb{E}[\Delta Y_i \mid X_i = x, A_i = 0]$$

The nonlinear $f(X)$ we built into the simulation

Trig + polynomial + log + interactions to generate the dataset. Nonlinearity is introduced throughout the simulation.

Propensity score

$$\text{logit } \mathbb{P}(D = 1 \mid X) = 0.5 \sin\left(\frac{\text{age}}{3}\right) + \frac{0.3 g^2}{10} - 0.4 \log(r + 1) + \frac{0.2 (\text{age} \cdot g)}{30}$$

Nonlinear treatment assignment

Pre-period outcome

$$Y_{\text{pre2}} = 3 \sin\left(\frac{\text{age}}{5}\right) + 0.7 g^{1.5} - \frac{0.4 r^2}{5} + 0.5 \text{age} + 30 + \varepsilon$$

Nonlinear baseline outcomes

Post-period shock

$$Y_{\text{post}} = Y_{\text{pre2}} + \tau D + 5 \sin\left(\frac{\text{age}}{3}\right) + \varepsilon$$

True treatment effect is set to $\tau = 1.5$, which serves as the ground truth.

Four DID methods

M1 Unconditional DID

WHAT IT DOES Basically unconditional

KEY ASSUMPTION Marginal parallel trends

FAILS WHEN Treated and control groups have different untreated trends

M2 Conditional Linear DID

WHAT IT DOES Models linearly using controls

KEY ASSUMPTION Linear conditional parallel trends

FAILS WHEN Nonlinear

M3 Propensity-Score DID (IPW)

WHAT IT DOES Uses propensity-score weights

KEY ASSUMPTION Correct propensity score

FAILS WHEN Poor common support or misspecified propensity score

M4 Nonlinear Conditional DID (GAM)

WHAT IT DOES Models flexibly using a GAM fit on controls

KEY ASSUMPTION Nonparametric conditional parallel trends

FAILS WHEN The control trend function is poorly estimated or poor controls

Unconditional DID

$$ETT_{\text{naive}} = (\bar{Y}_{\text{post}|D=1} - \bar{Y}_{\text{pre}|D=1}) - (\bar{Y}_{\text{post}|D=0} - \bar{Y}_{\text{pre}|D=0})$$

Difference each group's pre - post change, then take the difference of those differences.

Strengths

- Simplest possible — four sample means, no model
- Fully transparent and easy to communicate

Weaknesses

- Assumes parallel trends marginally
- No protection against compositional differences between treated and control
- Biased when untreated trends differ because groups differ in X

Linear Conditional DID (OLS outcome regression)

$$\theta_{ETT} = \mathbb{E}[Y_2 - Y_1 \mid D = 1] - \mathbb{E}[m_0(X) \mid D = 1]. \quad m_0(X) = \mathbb{E}[Y_2 - Y_1 \mid D = 0, X].$$

Fit among controls, predict the untreated counterfactual trend for treated units, then subtract.

Strengths

- Adjusts for observed covariates with Ordinary Least Squares (OLS)
- Closed-form, fast, and easy to interpret coefficients
- Often a meaningful improvement over the naive

Weaknesses

- Biased if true model is nonlinear
- Linearity is a strong restriction unless curvature/interactions are added

Propensity-Score DID (IPW) — Abadie (2005)

$$ETT_{IPW} = E \left[\frac{\Delta Y_i}{P(D_i = 1)} \cdot \frac{D_i - \pi_0(X_i)}{1 - \pi_0(X_i)} \right], \quad \pi_0(X_i) = P(D_i = 1 \mid X_i)$$

D is treatment; X are baseline covariates; π is the propensity score. It applied with inverse probability weighting (IPW).
propensity score = predicted probability of being treated given observed covariates.

Strengths

- Only the propensity score must be correct — no outcome model needed
- The weights track the true nonlinear selection

Weaknesses

- Sensitive when score is near 1 for treated-like controls
- Requires positivity/common support
- Can have high variance when weights are extreme

Our estimated propensity score is: $\hat{p}(X_i) = \hat{P}(D_i = 1 \mid \text{age}_i, \text{genre}_i, \text{release}_i)$

Nonlinear Conditional DID (GAM)

$$\widehat{ETT}_{GAM} = \widehat{\mathbb{E}}[Y_2 - Y_1 \mid D = 1] - \widehat{\mathbb{E}}[\widehat{m}_0^{GAM}(X) \mid D = 1], \quad \widehat{m}_0^{GAM}(X) = \widehat{\alpha} + s(\text{age}) + \gamma_{\text{genre}} + \gamma_{\text{release}}.$$

Replace OLS with a GAM fit on controls to estimate E flexibly.

Strengths

- Captures nonlinear untreated trends related to baseline covariates

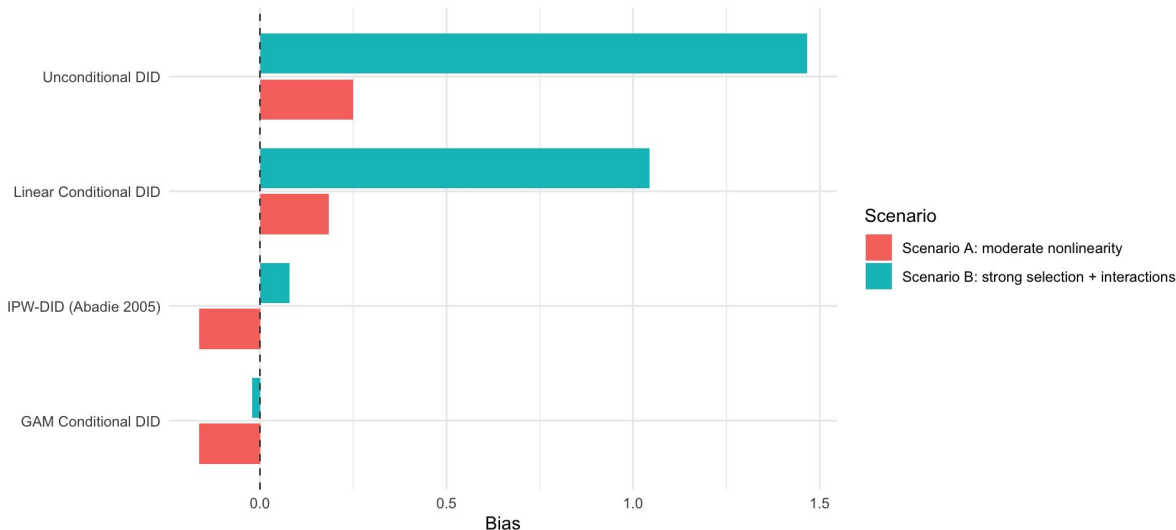
Weaknesses

- Relies on estimating the control trend function
- Smoothing choices can add variance and tuning sensitivity

GAM handles the tough nonlinear case best

Bias Comparison Across Two Nonlinear DID Scenarios

Scenario B is constructed to make unconditional DID much less reliable



Bias = Estimate - true ETT

$$\text{Bias} = \hat{\theta} - \tau$$

Naive DID has poor performance

+1.47 bias under strong selection

Linear DID is not a fix

Bias drops to +1.04. Cannot model nonlinear

IPW has good behavior

Significant lower bias comparing with naive and linear

GAM has lowest bias

Behaviors best in both scenario

Repository — code, data, results

Everything in this study is in our project repo.

CSV

synthetic_data.csv

The synthetic dataset (n = 2,000) — viewer-level age, genre, release, treatment, and minutes viewed at three timepoints.

PDF

Nonlinear DID: Method & Theory Extension.pdf

Full results report — DAG, theory, identification proofs, Scenario A & B tables, and bias figures.

R

simulate_data.R

Functions to generate synthetic data for different parameter settings (n, propensity coefficients, trend nonlinearity, true T).

R

run_models.R

Implements all four estimators (Naive, Linear, IPW, GAM) and the bias-comparison figure.

Thank you